

ORIGINAL ARTICLE

Open Access



Surface hydrophobicity-hydrophilicity switching induced interface heat and water transfer enhancement for high-efficiency solar steam generation

Yubo Tan¹, Hui Jin¹, Samuel S. Mao² and Shaohua Shen^{1*} 

Abstract

Beyond photothermal conversion, the surface wettability of light-absorbing materials should be also determinative to the efficiency of solar-driven interfacial steam generation (SISG). Herein, by modifying hydrophobic Cu nanoparticles (NPs) with a hydrophilic carbon (C) shell, hydrophilic Cu@C core-shell NPs were successfully fabricated and used for constructing evaporation films for SISG. In comparison to the film constructed with Cu NPs, the evaporation films constructed with Cu@C core-shell NPs exhibit much increased SISG efficiency, reaching 94.6% as high. Except for the localized surface plasmon resonance (LSPR) effect of Cu NPs ensuring the excellent photothermal conversion, it is experimentally and theoretically revealed that the surface wettability switching from hydrophobicity to hydrophilicity, as induced by C coating, is beneficial to heat transfer at the solid/liquid interface and water transport at the evaporative surface, thus improving the thermal-evaporation conversion performance for efficient SISG. However, the further thickened C shells would weaken the LSPR effect and hinder the interface heat and water transfer, leading to the decreased photothermal and thermal-evaporation conversion efficiencies, and thus the lowered SISG performances. This demonstration gives an alternative and promising access to the rational design of photothermal materials featured with switchable surface wettability ensuring interface heat and water transfer enhancement for efficient SISG.

Highlights

- Cu@C core-shell NPs with hydrophilic surface were successfully obtained via hydrothermal carbonization.
- The C shells induced surface wettability switching from hydrophobicity to hydrophilicity benefited heat and water transfer.
- The Cu@C core-shell NPs exhibited a high SISG efficiency reaching 94.6% under 1 sun illumination.

Keywords Surface wettability, Interface heat transfer, Photothermal conversion, Thermal-evaporation conversion, Solar-driven interfacial steam generation

*Correspondence:

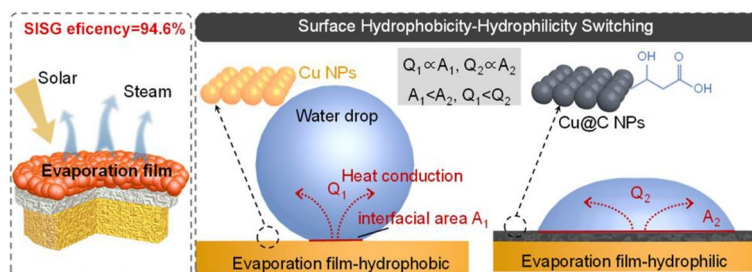
Shaohua Shen

shshen_xjtu@mail.xjtu.edu.cn

Full list of author information is available at the end of the article

Graphical Abstract

With surface wettability of Cu nanoparticles switched from hydrophobicity to hydrophilicity by the coating of C shells, the obtained Cu@C core-shell NPs exhibit much increased SISG efficiency, reaching 94.6% as high, synergistically attributed to the excellent photothermal conversion, the strengthened heat transfer at the solid/liquid interface and the enhanced water transport at the evaporative surface.



1 Introduction

Solar-driven interfacial steam generation (SISG) that could directly and efficiently utilize solar energy for water distillation and purification has been believed as one of promising and sustainable routes to mitigate the global shortage of fresh water, given its advantages of low cost, energy saving and zero CO₂ emission [1–3]. For a typical SISG process, solar light is harvested by light absorber and converted into heat at the evaporative surface, then driving the evaporation process with continuous water supply [1, 4]. Hence, in the past decades, great efforts have been carried out to improve the light harvest [5] and photothermal conversion abilities [6] as well as to enhance the thermal-evaporation and water transport processes [4, 7, 8], with a substantial improvement in SISG efficiency, increasing from only 30% to >90% under 1 sun illumination [1].

Undoubtedly, the light-absorbing materials, at which water is transported and evaporated, are the most important component determinative to the SISG performances. Till now, various materials, including plasmonic metals, semiconductors, carbonaceous materials, and polymers, etc. [9], with broadband solar light absorption and excellent photothermal conversion ability, have been developed for SISG [4, 10, 11]. Among these photothermal materials, some noble metals, such as Au and Ag, featured with local surface plasmon resonance (LSPR) effect, have received extensive exploration in SISG, due to their extraordinary talents of stable chemical properties and tunable light absorption enabled by size and morphology controlling [6, 12]. Halas et al. for the first time used hollow Au nanoparticles (NPs) to fabricate a photothermal evaporation device, with broadband light-absorption and strong photothermal ability, which was thus endowed with SISG efficiency over 80% [13]. Zhu

et al. constructed a plasmonic absorber with various-sized Au NPs self-assembled on a nanoporous alumina template for steam generation, which realized excellent SISG performance (~90% under 4 kW m⁻² solar irradiation), owing to the high light absorption efficiency (99%) and the significantly localized heating effect enabled by the hybridized LSPR effects between Au NPs and strong field enhancement in porous template [11]. Beyond that, the thermal-evaporation process, related to the heat transfer and water transport at the solid/liquid interface, which should be also essential to the SISG performance. Thus, plasmonic metal-based light absorbers with rationally engineered surface and interface structures have been designed to enhance interfacial heat transfer and water transport to realize high performance SISG [14]. For example, Liu et al. demonstrated a novel porous wooden flower decorated with hydrophilic plasmonic Ag-polydopamine core-shell NPs for SIGS with efficiency reaching 97.0% as high, which was believed to be attributed to the improved heat convection and water transport abilities, except for the excellent light absorption efficiency (98.7%) [15]. Despite of significant progress achieved by plasmonic noble metals, it is still a great challenge for large-scale applications in SISG due to their scarce reserves and high prices.

Alternatively, as an abundant and low-cost metal, Cu has been widely used for super-hydrophobic self-cleaning, condensation heat transfer enhancement [16, 17], and electrocatalysis, etc. [18, 19]. Although the LSPR effect endows Cu NPs as a promising candidate of photothermal material with significant resonance absorption in visible light region [20, 21], Cu NPs are still limited in SISG application, given the challenge in their morphological control and batch synthesis [22]. Moreover, Cu NPs featured with surface hydrophobicity [22, 23], giving rise

to the retarded heat transfer from Cu surface to water and the sluggish water transport at surface for interfacial thermal evaporation, always exhibit lower water evaporation rate than hydrophilic ones [24]. It could be then expected that by switching surface wettability from hydrophobicity to hydrophilicity, Cu NPs with controllable morphological and surface properties would be effective to photothermal conversion as well as thermal-evaporation, achieving high-efficiency SISG.

Herein, Cu NPs synthesized via an aqueous reduction process were coated with carbon (C) shells via a hydrothermal carbonization method using glucose as carbon precursor. In comparison to Cu NPs with SISG efficiency of only 59.5%, the obtained Cu@C core-shell NPs exhibit a remarkable increase in SISG efficiency reaching as high as 94.6% under 1 sun illumination, which stands at the highest level of non-noble metal based photothermal materials, and even higher than most of noble metals. It is experimentally and theoretically demonstrated that along with the satisfying photothermal conversion performance of plasmonic Cu NPs, the strengthened

heat transfer at solid/liquid interface and the rapid water transport at surface could be realized by the C coating induced surface wettability switching from hydrophobicity for Cu NPs to hydrophilicity for Cu@C core-shell NPs. Thus, the resulted enhancement in thermal-evaporation conversion greatly contributes to the much improved SISG performance. This study demonstrates a facile and effective strategy to design novel nanostructures for high efficiency SISG and related solar energy conversion, with switched surface wettability for interface heat and mass transfer enhancement.

2 Results and Discussion

In this study, Cu@C core-shell nanoparticles (NPs) were obtained by a two-step process, as schemed in Fig. 1a. Cu NPs were firstly synthesized via aqueous reduction, which were then coated by carbon (C) shells with glucose as carbon precursor via hydrothermal carbonization. In the first aqueous reduction step, the obtained Cu NPs exhibit homogeneous size distribution with average diameter of 80–100 nm (Fig. 1b).

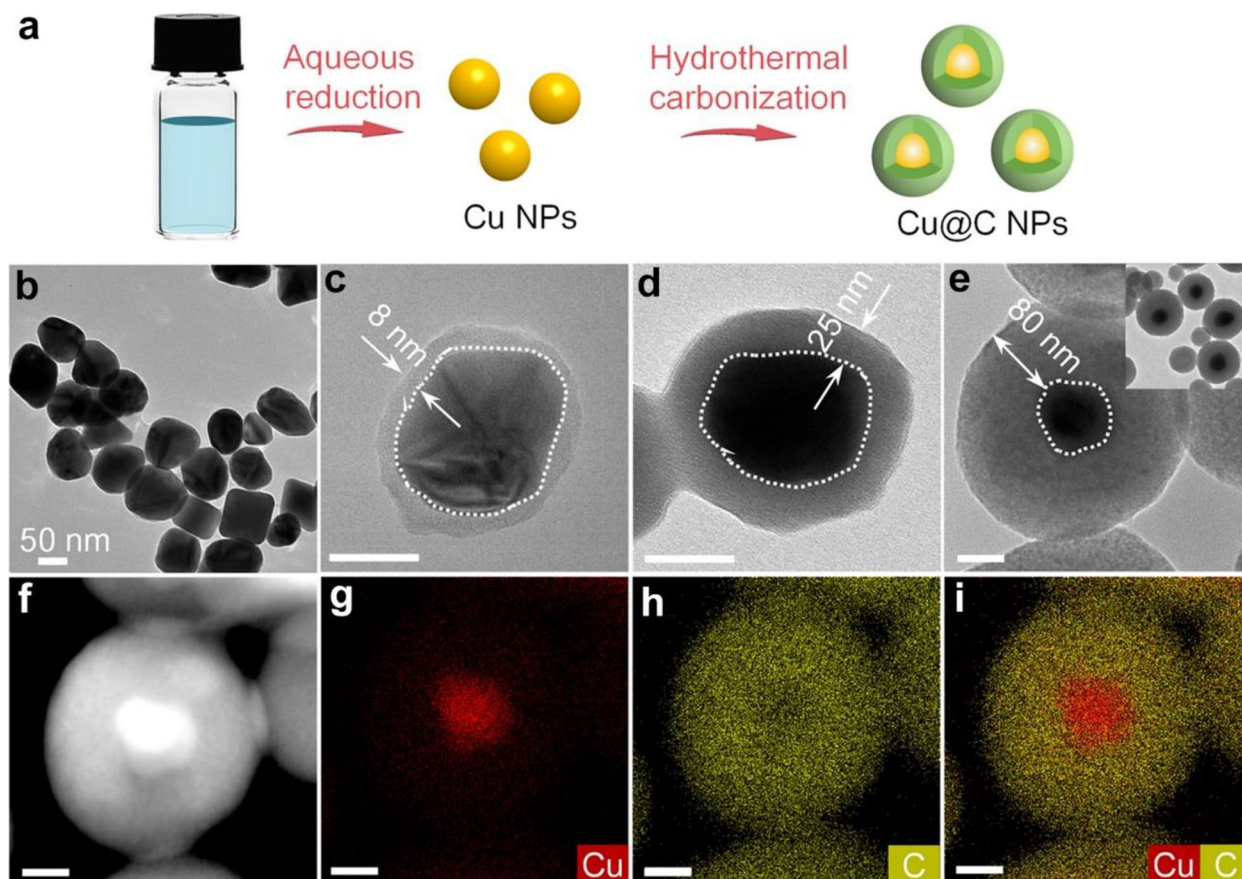


Fig. 1 **a** Schematic illustration of the synthesis of Cu@C core-shell NPs via a two-step method. TEM images of **b** Cu NPs and **c-e** Cu@C core-shell NPs with C shell thickness controlled at 8 nm (Cu@C-8), (25 nm) (Cu@C-25), and 80 nm (Cu@C-80), respectively. **f** Scanning TEM image and **g-i** associated element mappings of Cu@C-80. Scar bar: 50 nm

Then, Cu@C core-shell NPs were fabricated through the hydrothermal carbonization step (Fig. 1a). By varying the feeding amounts of Cu NPs versus glucose as carbon precursor, the C shells could be uniformly coated onto Cu NPs (Fig. 1c-e), with thickness finely controlled at 8 nm (Cu@C-8), 25 nm (Cu@C-25) and 80 nm (Cu@C-80) by feeding 3, 2, and 1 batches of Cu NPs during the second step, respectively. Without any lattice fringe observed, the coated C shells exist in amorphous phase, which agrees well with the aromatic domains with graphitic structures of carbon spheres [25] synthesized via hydrothermal carbonization [4]. The transmission electron microscopy (TEM) image of Cu@C-80 and the associated elemental mappings also confirm the successful synthesis of Cu@C core-shell NPs, with Cu and C elements well distributed at the core and shell regions (Fig. 1f-i), respectively.

X-ray diffraction (XRD) patterns were collected to evidence the successful synthesis of Cu NPs and then the uniform coating of C shells. As shown in Fig. 2a, the obtained Cu NPs present intensive diffraction peaks at $2\theta = 43.2^\circ$, 50.4° , and 74.1° , assigned to the (111), (200), and (220) planes of well crystallized zero-valent copper (#99-0034), respectively. In comparison,

with C shells coated, the XRD peaks related to Cu NPs are significantly weakened for Cu@C-8, and even hardly observed for Cu@C-25 and Cu@C-80, with the increasing thickness of the coated C shells. The disappearance of diffraction characteristics implies that C shells are well coated onto Cu NPs, which might block X-ray detection. In addition, one would note a broad peak at $2\theta = 20\text{--}28^\circ$ for all the Cu@C core-shell NPs, which suggests the amorphous phase of the coated C shells, as previously revealed by TEM images. It has been well recognized that the optical absorption ability of the light-absorbing materials is determinative to the photothermal efficiency and then the solar-driven interfacial steam generation (SISG) performance [2, 6, 26]. As shown in Fig. 2b, both Cu NPs and Cu@C core-shell NPs present excellent light absorption ability in visible light and even infrared light (>750 nm), well covering the wide-range solar spectrum. Specifically, Cu NPs exhibit excellent optical absorption in visible light, and especially an obvious absorption peak is easily observed at ca. 590 nm, which should be attributed to the LSPR effect of Cu NPs [20, 27]. Meanwhile, one would also note a broad optical absorption peak located at ca. 1150 nm. This wide-range optical absorption in

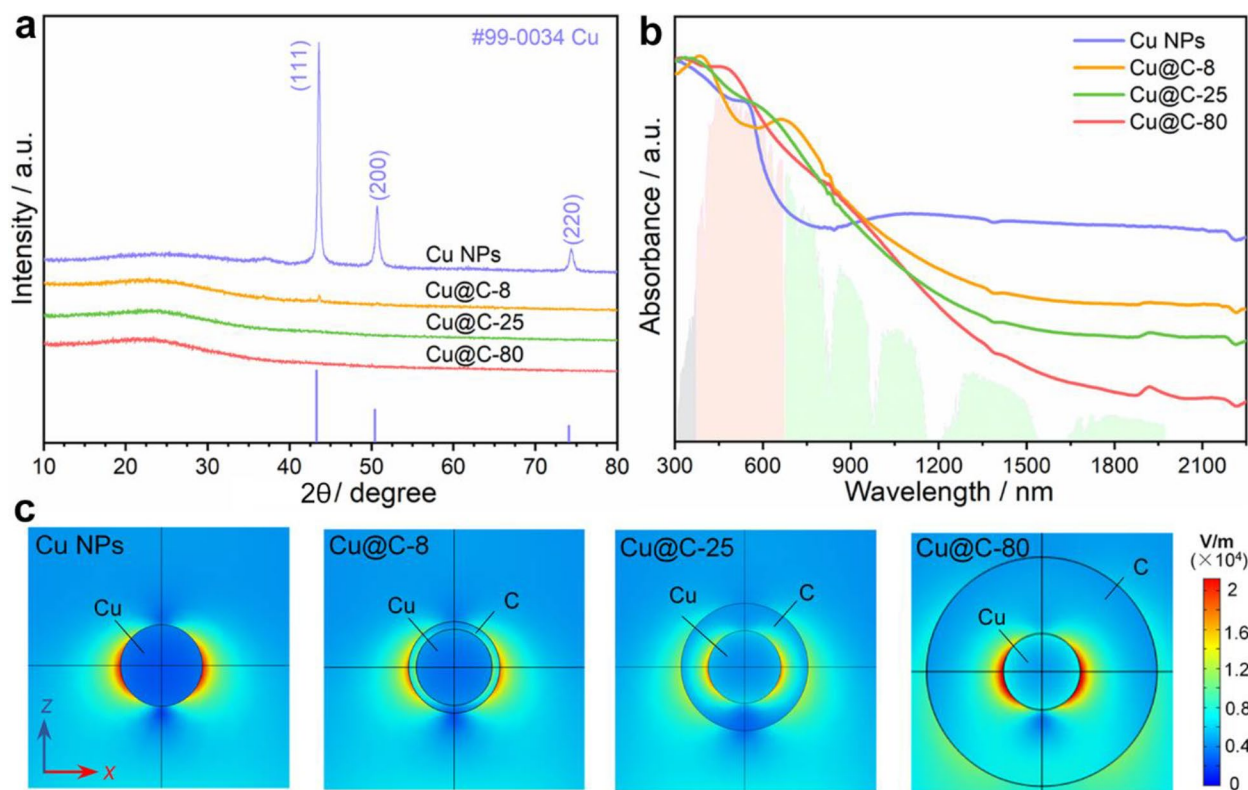


Fig. 2 **a** XRD patterns, and **b** UV-Vis-NIR spectra of Cu NPs and Cu@C core-shell NPs, **c** Spatial distribution of electromagnetic fields of Cu NPs and Cu@C core-shell NPs simulated by COMSOL Multiphysics

near-infrared region is believed to be induced by the high density of hybridized LSPR between the neighboring Cu NPs [11, 28]. With C shells coated, both the peaks at ca. 590 nm and ca. 1150 nm are remarkably weakened, owing to the C shells inhibiting the LSPR effects of Cu NPs. Especially, depending on the increasing C shell thickness, the optical absorption in the near-infrared region (ca. 1150 nm) is gradually reduced, given the thickened C shells separating the neighboring Cu NPs and thus eliminating the hybridized LSPR effect that induces the electromagnetic field coupling [6, 28, 29]. In addition, as compared to Cu NPs, the enhanced and extended optical absorption in visible light region (< 900 nm) is observable for all the Cu@C core-shell NPs, attributed to the excellent optical absorption of amorphous C shells derived from the hydrothermal carbonization of glucose. These results indicate that the coating of C shells onto Cu NPs would be beneficial to photothermal conversion and then SISG. The LSPR effect of Cu@C core-shell NPs depending on the C shells thickness was further validated by the spatial distribution of electromagnetic fields simulated with COMSOL Multiphysics, based on the grid independence study on optical simulations (Fig. S1). As shown in Fig. 2c, under solar light irradiation, Cu NPs display the intensive localized electromagnetic field at surface, induced by the LSPR effect. However, the electromagnetic field intensity is gradually decreased at the surface of Cu@C core-shell NPs, with the increasing thickness of the coated C shells. This comparative result again reveals that the coated C shells would weaken the hybridized LSPR effect of Cu NPs, corresponding well with the C shell thickness dependent optical absorption profiles of Cu@C core-shell NPs in the near-infrared region. Interestingly, the electromagnetic field intensity at the Cu/C interface is gradually strengthened, depending on the increasing C shell thickness. This phenomenon could be explained by the photon resonance accompanying with the LSPR effect of Cu NPs. With C shells coated, the incident photons would experience multiple scattering or refraction at the interface of Cu core and C shell, giving rise to the enhanced localized electromagnetic field [1]. Moreover, the thickening C shells could further confine the incident photons at the interfacial region by intensifying their scattering or refraction. Thus, the most intensive localized electromagnetic field is achieved at the Cu/C interfacial region of Cu@C-80 NPs.

All the above results and analysis verify that both Cu NPs and Cu@C core-shell NPs exhibit excellent optical absorption properties with optical profiles well matching the solar spectrum. Given their excellent optical properties benefiting photothermal conversion for effective

SISG, evaporation films were constructed with Cu NPs or Cu@C core-shell NPs deposited on Whatman paper as support (Fig. 3a). As shown in Fig. 3b, in the evaporation film constructed with Cu NPs serving as the photothermal conversion material, the layer of deposited Cu NPs and the Whatman paper are ca. 80 and ca. 350 μm in thick (Fig. 3b, left), respectively. It is clearly observed that Whatman paper is well covered by Cu NPs, which are mainly deposited at the top-surface of Whatman paper without leaking into the bottom region (Fig. 3b, right). For the evaporation films constructed with Cu@C core-shell NPs, one could also observe that Cu@C core-shell NPs are well deposited at the top-surface of Whatman paper (Fig. 3c). However, the thickness of the Cu@C core-shell NPs layer is reduced to be ca. 50 μm (Fig. 3c, left), and moreover, some Cu@C core-shell NPs are leaked into the bottom region of the Whatman paper (Fig. 3c, right). This very different penetration abilities of Cu NPs and Cu@C core-shell NPs into porous Whatman paper should be related to their varied surface wettability, which is switched from hydrophobicity to hydrophilicity with the coating of C shells, as further evidenced by water dropping test on the surface of the Cu and Cu@C evaporation films. As shown in Fig. 3d, with a water droplet dropped onto the surface, the Cu evaporation film exhibits the typical “lotus effect” (Fig. 3d, top) with the contact angle reaching 154.89° ($> 150^\circ$) (Fig. 3e), revealing the super-hydrophobic surface for the Cu evaporation film. Interestingly, for the Cu@C evaporation film, the water droplet would rapidly permeate into the film (Fig. 3d, bottom), which should be related to hydrophilic surface of the coated C shells enriched with C–OH and O–C=O groups (Fig. S2), ensuring the effective water transport by capillary infiltration (Fig. 3f) [3]. However, depending on the increasing thickness of the coated C shells, the water permeation times are significantly prolonged from 35.7 ms for Cu@C-8 to 105.7 ms for Cu@C-80. It is explainable that the thickened C shells would increase capillary resistance and then hinder water transport in the evaporation films.

The photothermal conversion performances of Cu@C core-shell NPs were evaluated by recording the stabilized temperatures of dry evaporation films under simulated solar light (Fig. 4a, dry). As visually recorded by thermal infrared camera, all the dry evaporation films constructed with Cu NPs or Cu@C core-shell NPs exhibit robust photothermal conversion ability induced by the LSPR effect of Cu NPs, with stabilized temperatures much higher than that detected on the blank one (pure Whatman paper, 30.2°C). One would note that depending on the thickening C shells, the stabilized temperatures decrease from 101.3°C for Cu NPs to 91.7°C for Cu@C-80. This weakened photothermal conversion

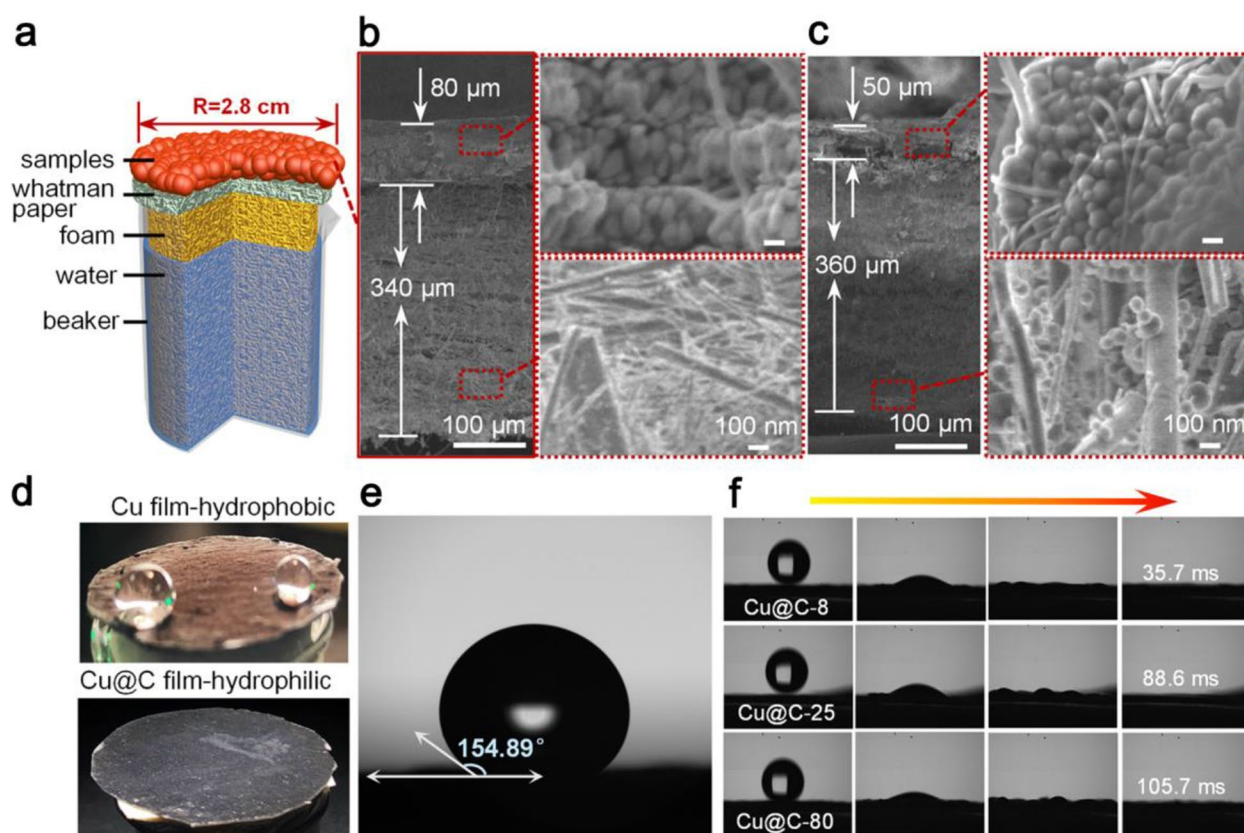


Fig. 3 **a** Schematic illustration of evaporation films (Whatman paper deposited with Cu NPs or Cu@C core-shell NPs for SISG). **b, c** SEM images of evaporation films constructed with Cu NPs and Cu@C core-shell NPs. **d** Digital images of evaporation films with dropping water. **e** Photograph of a water droplet at the surface of the evaporation film constructed with Cu NPs with contact angle measured. **f** High speed camera recorded water permeation processes

ability of the dry evaporation films could be explained by the decreases in the light absorption ability (Fig. 2b) and the LSPR effect (Fig. 2c) of Cu@C core-shell NPs with thickening C shells. The SISG performance was then preliminarily evaluated by recording the real-time temperatures of the evaporation films during water evaporation under the irradiation of simulated solar light. Given the temperatures increasing rapidly in the initial 10 s (Fig. 4b) and then reaching stabilization after 600 s (Fig. 4c) for all the wet evaporation films, the stabilized temperatures recorded at 1000 s are used for the subsequent heat transfer calculation, *i.e.*, 89.0 °C (Cu NPs), 47.5 °C (Cu@C-8), 48.0 °C (Cu@C-25), and 48.8 °C (Cu@C-80) (Fig. 4a, wet). It is clear that the stabilized temperatures of wet evaporation films are much lower than those of dry evaporation films (Fig. 4a, dry). The temperature drop (Tem. drop) between the dry and wet evaporation films reveals that the water thermal-evaporation conversion during SISG process would take the heat away and then cool the evaporation films. The very small Tem. drop (1.1 °C) observed for the Whatman paper (blank) indicates its negligible

water thermal-evaporation ability. In comparison to the Cu evaporation film with a small Tem. drop of 12.3 °C, all the Cu@C evaporation films exhibit much larger Tem. drops, due to the efficient water transport in the films ensured by the hydrophilic C shells. However, depending on the increasing C shell thickness, the Tem. drops are gradually decreased from 47.4 °C for Cu@C-8 to 42.9 °C for Cu@C-80, which implies that the thickened C shells would weaken water transport in the Cu@C evaporation films. It is further theoretically evidenced that the thickened C shells could also weaken the heat transfer from Cu cores to the surface of Cu@C core-shell NPs. As shown in Fig. 4d, under the simulated solar light irradiation, the thermal power density of the Cu core gradually increases with the coating and the further thickening of the C shells, indicating that the heat is localized at the surface of Cu core rather than transferred to the surface of Cu@C core-shell NPs. These theoretical results, corresponding well with the stabilized temperatures of dry evaporation films recorded under simulate solar irradiation, demonstrate that the poor thermal conductivity of

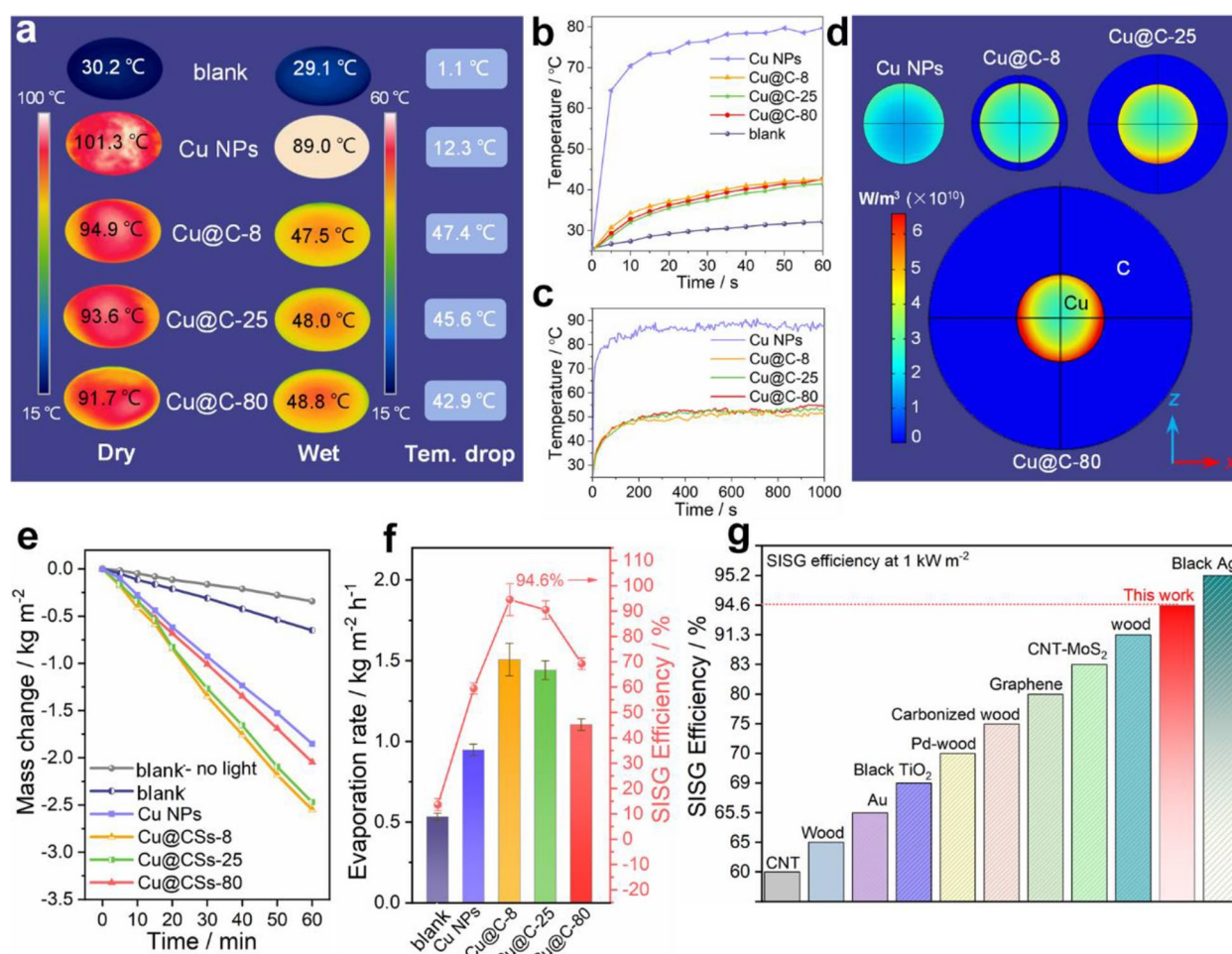


Fig. 4 **a** Thermal images of dry and wet evaporation films with temperatures stabilized under simulated solar light. Real-time temperatures of the evaporation films recorded by thermal infrared camera during SISG process for **b** 60 s and **c** 1000 s. **d** Calculated thermal power density distribution of Cu NPs and Cu@C core-shell NPs by COMSOL Multiphysics under simulated solar light. **e** Cumulative mass change for pure water evaporation, **f** Water evaporation rates, and SISG efficiencies of different evaporation films. **g** SISG efficiencies of typical materials reported in previous references

thickened C shells [3] and the weakened heat transfer at the Cu/C interface should be responsible to the reduced photothermal conversion ability of the Cu@C core-shell NPs coated with thickening C shells, as discussed in the following sections.

To evaluate the water thermal-evaporation ability of the evaporation films, the cumulative mass change of pure water during SISG process were recorded (Fig. S3). For the Whatman paper (blank), a very small amount of water is vaporized upon irradiation of simulated solar light for 60 min (Fig. 4e), with water evaporation rate calculated to be only 0.53 kg m⁻² h⁻² (Fig. 4f). Encouragingly, both the Cu and Cu@C evaporation films present significant increase in the cumulative mass change of pure water during SISG process, which should be related to the excellent photothermal conversion ability of Cu NPs contributing to the much improved SISG

performances. Notably, in comparison to the Cu evaporation film that presents a relatively low water evaporation rate (0.95 kg m⁻² h⁻²), all the Cu@C evaporation films have water evaporation rates much increased, especially, to be 1.51 kg m⁻² h⁻² as high for Cu@C-8. Given the super-hydrophobic surface of Cu NPs resulting in the poor water transport in the Cu evaporation film, such a great increase in water evaporation rates should be attributed to the surface wettability switching from hydrophobicity to hydrophilicity induced by C shell coating, contributing to the efficient water transport in the Cu@C evaporation films. However, depending on the thickening C shells coated onto Cu NPs, the recorded cumulative mass change during SISG process are gradually decreased for the Cu@C evaporation films (Fig. 4e), with the calculated water evaporation rates decreasing from 1.51 kg m⁻² h⁻² for Cu@C-8 as the maximum to

$1.10 \text{ kg m}^{-2} \text{ h}^{-2}$ for Cu@C-80 (Fig. 4f). These visualized and calculated results well support the previous inference that the thickening C shells would weaken the water transport ability in the evaporation films and the heat transfer ability at the Cu/C interface, as evidenced by the analysis of Tem. drop (Fig. 4a) and the thermal power density (Fig. 4d), respectively, which together gives rise to the decreased water thermal-evaporation ability. Based on the water evaporation rates, the SISG efficiency (η) of the evaporation films could be calculated by Eq. 1 [4, 30]:

$$\eta = \frac{\dot{m} \cdot H_m}{\dot{I} \cdot T} \quad (1)$$

where $\dot{m}$ is the water evaporation rate ($\text{kg m}^{-2} \text{ h}^{-1}$), H_m is the latent heat of evaporation set as 2260 kJ kg^{-1} , $\dot{I}$ is the power density of the incident simulated solar light set as 1000 W m^{-2} (1 sun illumination), T is the duration time of irradiation set as 1 h. As calculated from the water evaporation rates, it is not surprising that the SISG efficiencies of the Cu@C evaporation films are much higher than that of the Cu evaporation film, with the highest SISG efficiency achieved over the Cu@C-8 evaporation film, reaching 94.6% as high (Fig. 4f). Encouragingly, this value stands at the forefront of the previous results reported for non-noble metal based photothermal materials and even outperforms most of noble metal-based materials (Fig. 4g). Depending on the increasing thickness of C shells, the SISG efficiency of the Cu@C evaporation films gradually decreases, due to the decreasing photothermal and thermal-evaporation conversion abilities of the films with thickening C shells. To deserve to be mentioned, as compared to the Cu@C evaporation films, the Cu evaporation film possesses much better photothermal

conversion but exhibits much lower SISG efficiency. Such discrepancy could be explained by the super-hydrophobic surface of Cu NPs. Specifically, the poor wettability of Cu NPs minimizes the interfacial contact area between solid (Cu NPs) and liquid (water), which not only prevents the water transport in the Cu evaporation film, but also hinders the heat transfer through the Cu/water solid/liquid interface, therefore significantly reducing the heat utilization for SISG. These results reveal that, except for the photothermal conversion ability of evaporation films, the thermal-evaporation conversion performance depending on the surface wettability (*i.e.*, hydrophobicity and hydrophilicity) of light-absorbing materials should also determinatively contribute to the SISG efficiency of evaporation films. Moreover, the almost unchanged appearance (*i.e.*, color and surface texture) for the evaporation films before and after pure water evaporation (Fig. S4), confirming their good mechanical, thermal and chemical stability during SISG process.

In order to deeply understand how the surface wettability affects the thermal-evaporation process during SISG, the interface heat transfer processes were schematically proposed in the evaporation films constructed with hydrophobic Cu NPs and hydrophilic Cu@C core-shell NPs (Fig. 5a). For the evaporation film constructed with hydrophobic Cu NPs, the poor surface wettability endows the contact between the evaporation film and the water with very small interfacial area (A_1) for heat conduction from the film to the water for the subsequent thermal-evaporation conversion. In contrast, benefiting from the coating of C shells with plenty of C–OH and O–C=O surface groups [3], the evaporation films constructed with Cu@C core-shell NPs exhibit a hydrophilic surface

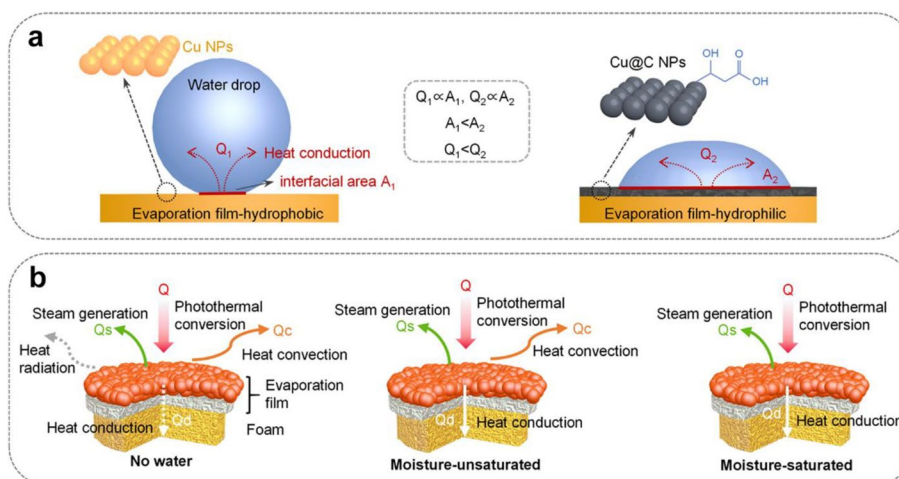


Fig. 5 **a** Proposed heat conduction models at the evaporation interface (contact area between the top surface of the evaporation film and water) of evaporation films. **b** Proposed heat transfer models at the evaporator region in the dry, moisture-unsaturated, and moisture-saturated evaporation films for SISG

with water well spreading on the films, forming much enlarged interfacial area (A_2) for the enhanced interfacial heat conduction for efficient thermal-evaporation.

Except for interface heat transfer, one should keep in mind that the water transport ability greatly relying on the surface wettability would also determine the thermal-evaporation process, given water as heat carrier being responsible to the heat transfer in evaporation region (including the evaporation film, air domain upon near-surface and foam below near-surface of the film). Note that the enhanced water transport ensures the quick supply of water to efficiently utilize the thermal for water evaporation. Here, three models of heat transfer at the evaporator region were proposed for the evaporation films with different water transport abilities, *i.e.*, dry evaporation film (Fig. 5b, No water), moisture-unsaturated evaporation film (Fig. 5b, Moisture-unsaturated), and moisture-saturated evaporation film (Fig. 5b, Moisture-saturated).

With no water supplied in the dry evaporation film, the heat transfer in the evaporator region mainly involves the heat input over photothermal conversion and the heat loss (including heat radiation from film to air, heat convection on the film surface, and heat conduction from film to foam), which should follow the heat balance described in Eqs. 2, 3, 4, 5 [31]:

$$Q = Q_c + Q_{rad} + Q_d \quad (2)$$

$$Q_c = h_c \cdot (T_d - T_\infty) \quad (3)$$

$$Q_{rad} = \varepsilon_f \cdot \left[\left(\frac{T_d}{100} \right)^4 - \left(\frac{T_\infty}{100} \right)^4 \right] \quad (4)$$

$$Q_d = k_0 \cdot (T_d - T_\infty) \quad (5)$$

where Q is the input heat flux by photothermal conversion of the evaporation films (W m^{-2}), Q_c is the heat convection flux (W m^{-2}), Q_{rad} is the heat radiation flux (W m^{-2}), Q_d is the heat conduction flux (W m^{-2}), h_c is the average heat transfer coefficient of the dry films ($\text{W m}^{-2} \text{K}^{-1}$), T_d is the stabilized temperature of the dry films recorded in Fig. 4a ($^\circ\text{C}$), T_∞ is the environment temperature set as 25°C , ε_f is the emissivity of the evaporation films, with the value of the Cu evaporation film set as 0.6 and the Cu@C evaporation films set as 0.9530, k_0 is the thermal conductivity of the foam, with the value usually smaller than $0.03 \text{ W m}^{-1} \text{K}^{-1}$ attributed to the excellent thermal insulation property of dry foam [32]. Hence, the value of Q_{rad} and Q_d is far less than 1 depending on the calculation, which could be ignorable in the model of the dry evaporation films. Consequently, the heat flux

calculation in the dry evaporation films could be simplified as the following Eq. 6 [4]:

$$Q = Q_c = h_c \cdot (T_d - T_\infty) \quad (6)$$

Herein, given the Cu NPs and Cu@C core-shell NPs with different C shells thickness possessing different surface and thermal insulation properties, the average heat transfer coefficient of the evaporation films (h_c) is varied. As calculated from Eqs. S1-S8 and Table S1 (see Supporting information), h_c of the evaporation films is determined to be $13.9 \text{ W m}^{-2} \text{K}^{-1}$ (Cu NPs), $13.6 \text{ W m}^{-2} \text{K}^{-1}$ (Cu@C-8), $13.6 \text{ W m}^{-2} \text{K}^{-1}$ (Cu@C-25), and $13.5 \text{ W m}^{-2} \text{K}^{-1}$ (Cu@C-80), respectively. With the calculated value of h_c introduced in Eq. 6, Q could be then calculated out and used for calculating the heat flux in other two heat transfer models in the evaporator region.

Unlike the simple heat transfer model in dry evaporation film, two different heat transfer models would exist in wet evaporation films with different water transport abilities. When water transport from the bulk water to the surface of the evaporation films is slower than water evaporation at the film surface, the water supply is insufficient for SISG and thus the evaporation films are unsaturated with moisture. In this case, the heat transfer in the evaporation region of the moisture-unsaturated films would be more complicated than that in the dry evaporation films. As described in Fig. 5b (Moisture-unsaturated), the input heat flux by photothermal conversion is not only utilized for steam generation but also lost during SISG, *i.e.*, heat loss, including heat conduction from the film to the bulk water and heat convection from the film to the air. Thus, the heat transfer in the evaporation region of the moisture-unsaturated films should satisfy the following Eqs. 7, 8 and 9 [4]:

$$Q = Q_s + Q_c + Q_d \quad (7)$$

$$Q_s = Q_s = \dot{m} \cdot H_m \quad (8)$$

$$Q_L = Q_C + Q_d \quad (9)$$

where Q_s is the heat flux of water evaporation (W m^{-2}), Q_L is the heat loss flux in the evaporator region. Then, Q_s and Q_L could be verified by combining Eqs. 7, 8, 9 and the previously calculated Q values.

It is understandable that, once the water transport rate is high enough to ensure adequate water supply for steam generation, the evaporation films would be saturated with moisture (Fig. 5b, Moisture-saturated). In this case, with the evaporation films fully covered by water, the heat convection flux could be negligible, and the heat loss only counts in the heat conduction from the evaporation films

to the bulk water. Then, the heat transfer in the evaporation region of the moisture-saturated films follows Eqs. 10, 11:

$$Q = Q_S + Q_d \quad (10)$$

$$Q_L = Q_d \quad (11)$$

As analyzed above, before calculating the heat fluxes based on the moisture-unsaturated or saturated model (Fig. 5b) proposed for the evaporator region of evaporation films during SISG, the water saturation of the films need to be verified by calculating the water capacity of the films. The water capacity could be confirmed by calculating the volume ratio of water absorbed in the evaporation films before and after the SISG process by following Eq. S9 (see Supporting information). As shown in Fig. 6a, the

water capacity of Cu@C-8 and Cu@C-25 evaporation films is calculated to be 0.78 and 0.83 before SISG, respectively, which well maintains at 0.78 and 0.81 after SISG, implying that the heat transfer in evaporator region of Cu@C-8 and Cu@C-25 evaporation films should be subjected to the moisture-saturated model. In comparison, the Cu@C-80 evaporation film should be subjected to the moisture-unsaturated model, as its water capacity decreases significantly from 0.82 before SISG to 0.76 after SISG. One should note that, although the water capacity of the Cu evaporation film keeps almost unchanged for SISG, its value of only 0.34 before SISG is much smaller than those of the Cu@C evaporation films. Such a small water capacity should be related to the super-hydrophobic surface of the Cu evaporation film, which would hinder the water

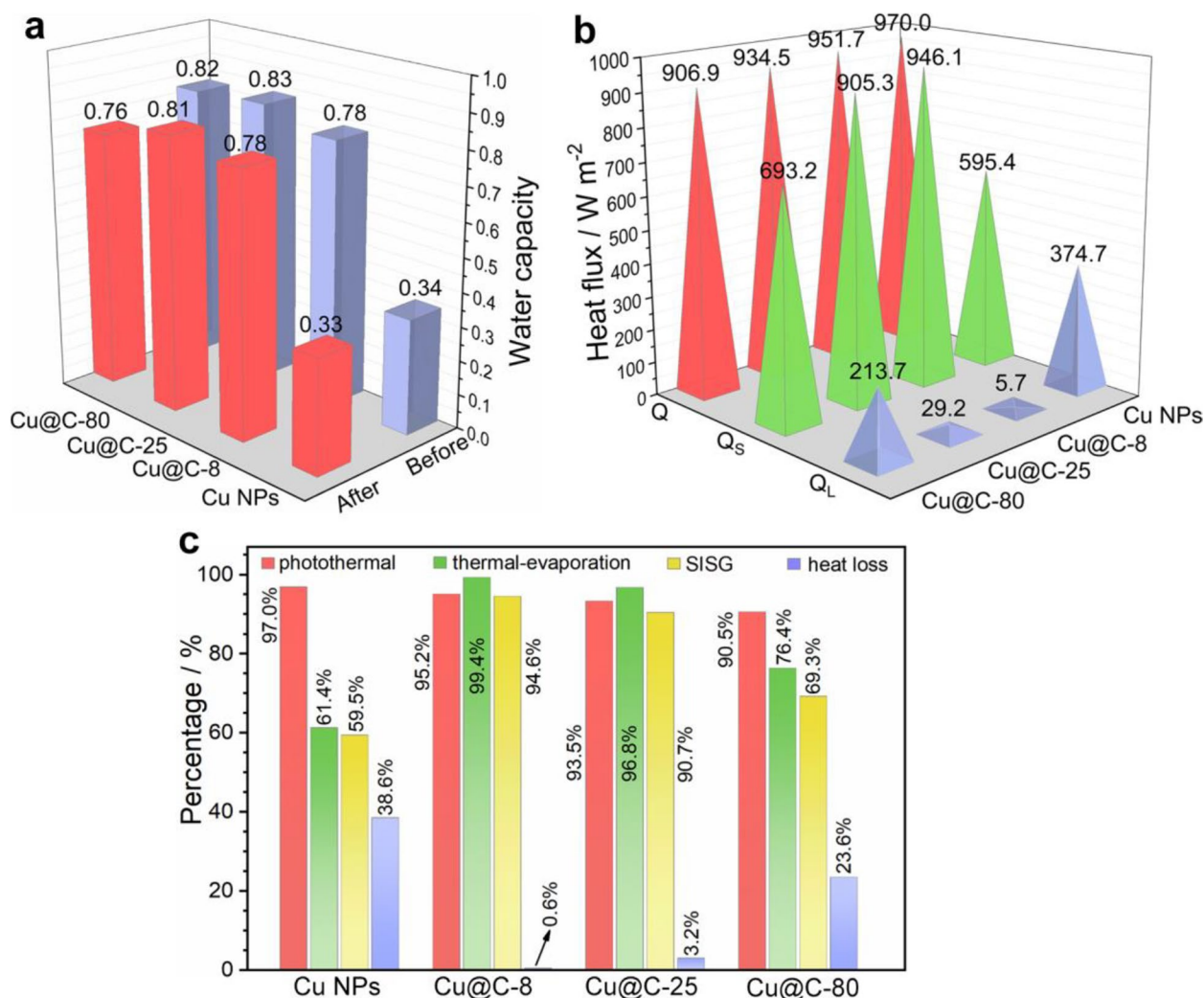


Fig. 6 **a** Water capacity of different evaporation films before and after SISG. **b** Calculated values of heat flux. **c** Photothermal efficiency, thermal-evaporation efficiency, SISG efficiency, and percentage of heat loss in the evaporation films constructed with Cu NPs and Cu@C core-shell NPs

absorption and transport, preventing the coverage of water on the evaporation film. Thus, the Cu evaporation film is also subjected to the moisture-unsaturated model.

Depending on the heat transfer models confirmed for the different evaporation films, the heat flux in the evaporator region could be calculated by following Eqs. 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12. As shown in Fig. 6b, the input heat flux by photothermal conversion (Q) decreases from 970.0 W m^{-2} for Cu NPs as the highest to 906.9 W m^{-2} for Cu@C-80, which should be related to the weakened light absorption (Fig. 2) and photothermal conversion (Fig. 4) abilities of the evaporation films, depending on the increasing thickness of the coated C shells. With the surface wettability switched from hydrophobicity to hydrophilicity by the coating of C shells, the heat flux of water evaporation (Q_s) significantly increases from 594.5 W m^{-2} for Cu NPs to 946.1 W m^{-2} for Cu@C-8, and then gradually decreases to be 693.2 W m^{-2} for Cu@C-80 with the C shell thickness increasing. This change in Q_s is opposite to that observed for the heat loss flux (Q_L), which significantly decreases from 374.7 W m^{-2} for Cu NPs to 5.7 W m^{-2} for Cu@C-8 with the coating of C shells, and then gradually increases to 213.7 W m^{-2} for Cu@C-80 with the thickening C shells. These comparative trends in Q_s and Q_L not only well evidence that the excessive heat loss at the super-hydrophobic surface should account for the poor SISG performance achieved for the Cu evaporation film (Fig. 4), but also corroborate that, except for the gradually decreasing photothermal conversion, the increasing heat loss at the hydrophilic surface could also weaken the SISG performance of Cu@C evaporation film with thickening C shells.

Based on the above calculation results, the energy conversion efficiency, including photothermal conversion efficiency (η_p), thermal-evaporation conversion efficiency (η_t), and heat loss efficiency (η_L), could be then evaluated by following Eqs. 12, 13, 14:

$$\eta_p = \frac{Q}{I} \quad (12)$$

$$\eta_t = \frac{Q_s}{Q} \quad (13)$$

$$\eta_L = \frac{Q_L}{Q} \quad (14)$$

As shown in Fig. 6c, all the evaporation films exhibit photothermal conversion efficiency above 90%, which is slightly decreased from 97.0% for the Cu evaporation film to 90.5% for the Cu@C-80 evaporation film, corresponding well with the reduced light absorption depending on the coating and the further thickening of C shells. Such a slight change reveals that the surface

hydrophobicity-hydrophilicity switching induced by C shell coating does not bring great impact on the photothermal conversion of the evaporation films. It is then observed that depending on the coating and the further thickening of C shells, the thermal-evaporation conversion efficiency increases significantly from 61.4% for Cu NPs to 99.4% for Cu@C-8 as the highest and then gradually decreases to 76.4% for Cu@C-80, while the heat loss efficiency decreases from 38.6% for Cu NPs to 0.6% for Cu@C-8 as the lowest and then gradually increases to 23.6% for Cu@C-80. As a result, the SISG efficiency increases from 59.5% for Cu NPs to 94.6% as the highest for Cu@C-8 and then decreases to 69.3% for Cu@C-80. It is thus suggested that beyond the excellent photothermal conversion ability, efficient thermal-evaporation conversion should greatly contribute to the improved SISG performance. Especially, in contrast to the Cu evaporation film, the Cu@C-8 evaporation film realizes a significant improvement in the thermal-evaporation conversion efficiency, which should be attributed to the effective water supply and interface heat transfer for the maximum utilization of thermal for water evaporation, benefitted from the C shell coating induced surface wettability switching from hydrophobicity to hydrophilicity.

3 Conclusions

In summary, starting from the aqueous reduction fabricated Cu NPs, Cu@C core-shell NPs with hydrophilic surface and tunable shell thickness were successfully obtained via a hydrothermal carbonization method. With the coating of C shells, Cu@C core-shell NPs exhibit remarkable increase in SISG efficiency, reaching 94.6% as high, which stands at the highest levels reported for the non-noble metal based photothermal materials and even outperforms most of noble metals.

It is experimentally and theoretically verified that the great improvement in SISG performance for Cu@C core-shell NPs should be attributed to: 1) the excellent photothermal conversion ability of plasmonic Cu NPs; 2) the C coating induced surface hydrophobicity-hydrophilicity switching that enhances the heat transfer at the solid/liquid interface and the water transport at the evaporative surface, thus realizing the efficient utilization of the photothermal heat for water evaporation. However, further thickening of C shells would weaken the photothermal conversion, the heat transfer at the solid/liquid interface, and the water transport at the evaporative surface, which together results in the decreased SISG efficiencies.

This study demonstrates a facile and effective approach to surface wettability engineering that synergizes the interfacial heat and mass transfer

enhancement for efficient SISG, which would enrich the rational design of novel nanostructures for efficient solar energy conversion and utilization.

Abbreviations

SISG	Solar-driven interfacial steam generation
NPs	Nanoparticles
C	Carbon
LSPR	Localized surface plasmon resonance
XRD	X-ray diffraction
TEM	The transmission electron microscopy
Tem. drop	Temperature drop

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s43979-023-00051-x>.

Additional file 1. Experimental methods with materials and characterization. Calculation methods with confirmation of average heat transfer coefficient and water capacity of the evaporation films. Optical simulations (light absorption and photothermal power density distribution) on Cu NPs and Cu@C core-shell NPs; Grid independence study on optical simulations for a single Cu@C-80 core-shell nanoparticle (**Fig. S1**); FTIR spectra of Cu@C core-shell NPs (**Fig. S2**); Schematic diagram of test system for SISG (**Fig. S3**). Photos of evaporation films constructed with Cu NPs and Cu@C core-shell NPs before and after pure water evaporation (**Fig. S4**). The calculation parameters involved in the confirmation of average heat transfer (**Table S1**).

Acknowledgements

The authors thank the financial support from the National Natural Science Foundation of China (52225606, 51888103), the Natural Science Basic Research Program of Shaanxi Province (2019JCW-10), the "Fundamental Research Funds for the Central Universities", and "The Youth Innovation Team of Shaanxi Universities".

Authors' contributions

For author contributions, YT was in charge of the experimental design and analysis, synthesis and characterization of the samples, and original draft writing. HJ conceived the idea and assisted with the simulations and calculations. SM helped with simulations and calculations. SS administrated the project, supervised the experiments, wrote and revised the draft, and raised funding. The final version of the paper has been approved by all authors.

Funding

Open access funding provided by Shanghai Jiao Tong University. This work was supported by the National Natural Science Foundation of China (52225606, 51888103), the Natural Science Basic Research Program of Shaanxi Province (2019JCW-10), the "Fundamental Research Funds for the Central Universities", and "The Youth Innovation Team of Shaanxi Universities".

Availability of data and materials

The data and materials are available upon reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

All authors agree to the publication of this manuscript.

Competing interests

Shaohua Shen is an editorial board member for Carbon Neutrality and was not involved in the editorial review, or the decision to publish this article. All authors declare that there are no competing interests.

Author details

¹International Research Center for Renewable Energy, State Key Laboratory of Multiphase Flow in Power Engineering, Xi'an Jiaotong University, Shaanxi 710049, China. ²Department of Mechanical Engineering, University of California at Berkeley, Berkeley CA94720, USA.

Received: 13 February 2023 Revised: 19 April 2023 Accepted: 23 April 2023

Published online: 03 May 2023

References

- Chen C, Kuang Y, Hu L (2019) Challenges and opportunities for solar evaporation. *Joule* 3(3):683–718. <https://doi.org/10.1016/j.joule.2018.12.023>
- Zhao Y, Gao W, Li S, Williams G, Mahadi A, Ma D (2019) Solar-versus thermal-driven catalysis for energy conversion. *Joule* 3(4):920–937. <https://doi.org/10.1016/j.joule.2019.03.003>
- Xu Q, Niu Y, Li J, Yang Z, Gao J, Ding L, Ni H, Zhu P, Liu Y, Tang Y, Lv Z, Peng B, Hu T, Zhou H, Xu C (2022) Recent progress of quantum dots for energy storage applications. *Carbon Neutrality* 1:13. <https://doi.org/10.1007/s43979-022-00002-y>
- Tan Y, Liu M, Wei D, Ren J, Wang Y, Jin H, Shen S (2021) Oriented thermal etching of hollow carbon spheres with delicate heat management for efficient solar steam generation. *Int J Heat Mass Tran* 178:121579. <https://doi.org/10.1016/j.ijheatmasstransfer.2021.121579>
- Wu L, Zhao B, Gao D, Jiao D, Hu M, Pei G (2023) Solar transparent and thermally insulated silica aerogel for efficiency improvement of photovoltaic/thermal collectors. *Carbon Neutrality* 2:6. <https://doi.org/10.1007/s43979-023-00046-8>
- Chen J, Feng J, Li Z, Xu P, Wang X, Yin W, Wang M, Ge X, Yin Y (2019) Space-confined seeded growth of black silver nanostructures for solar steam generation. *Nano Lett* 19(1):400–407. <https://doi.org/10.1021/acs.nanolett.8b04157>
- Zhao H, Huang J, Zhou J, Chen L, Wang C, Bai Y, Zhou J, Deng Y, Dong W, Li Y, Yu S (2022) Biomimetic design of macroporous 3D truss materials for efficient interfacial solar steam generation. *ACS Nano* 16(3):3554–3562. <https://doi.org/10.1021/acsnano.1c10184>
- Liu Y, Yu S, Feng R, Bernard A, Liu Y, Zhang Y, Duan H, Shang W, Tao P, Deng SC, T. (2015) A bioinspired reusable paper-based system for high-performance large-scale evaporation. *Adv Mater* 27(17):2768–2774. <https://doi.org/10.1002/adma.201500135>
- Lin T, Yang C, Wang Z, Yin H, Lu X, Huang F, Lin J, Xie X, Jiang M (2014) Effective nonmetal incorporation in black titania with enhanced solar energy utilization. *Energy Environ Sci* 7:967–972. <https://doi.org/10.1039/C3EE42708K>
- Hao D, Yang Y, Xu B, Cai Z (2018) Efficient solar water vapor generation enabled by water-absorbing polypyrrole coated cotton fabric with enhanced heat localization. *Appl Therm Eng* 141:406–412. <https://doi.org/10.1016/j.applthermaleng.2018.05.117>
- Zhou L, Tan Y, Ji D X, Zhu B, Zhang P, Xu J, Gan Q, Yu Z, Zhu J (2016) Self-assembly of highly efficient broadband plasmonic absorbers for solar steam generation. *Sci Adv* 2(4):e1501227. <https://doi.org/10.1126/sciadv.1501227>
- Fuzil N, Othman N, Alias N, Marpani F, Othman M, Ismail A, Lau W, Li K, Kusworo T, Ichinose I, Shirazi, M (2021) A review on photothermal material and its usage in the development of photothermal membrane for sustainable clean water production. *Desalination* 517:115259. <https://doi.org/10.1016/j.desal.2021.115259>
- Neumann O, Feronti C, Neumann A, Dong A, Schell K, Lu B, Kim E, Quinn M, Thompson S, Grady N, Nordlander P, Oden M, Halas N (2013) Compact solar autoclave based on steam generation using broadband light-harvesting nanoparticles. *Proc Natl Acad Sci USA* 110(29):11677–11681. <https://doi.org/10.1073/pnas.1310131110>
- Zolotavin P, Alabastri A, Nordlander P, Natelson D (2016) Plasmonic heating in au nanowires at low temperatures: the role of thermal boundary resistance. *ACS Nano* 10(7):6972–6979. <https://doi.org/10.1021/acsnano.6b02911>
- Chen S, Sun Z, Xiang W, Shen C, Wang Z, Jia X, Sun J, Liu C (2020) Plasmonic wooden flower for highly efficient solar vapor generation. *Nano Energy* 76:104998. <https://doi.org/10.1016/j.nanoen.2020.104998>

16. Wen R, Li Q, Wu J, Wu G, Wang W, Chen Y, Ma X, Zhao D, Yang R (2017) Hydrophobic copper nanowires for enhancing condensation heat transfer. *Nano Energy* 33:177–183. <https://doi.org/10.1016/j.joule.2017.11.010>
17. Miljkovic N, Enright R, Nam Y, Lopez K, Dou N, Sack J, Wang E (2013) Jumping-droplet-enhanced condensation on scalable superhydrophobic nanostructured surfaces. *Nano Lett* 13(1):179–187. <https://doi.org/10.1021/nl303835d>
18. Chen S, Wang B, Zhu J, Wang L, Ou H, Zhang Z, Liang X, Zheng L, Zhou L, Su Y, Wang D, Li Y (2021) Lewis acid site-promoted single-atomic Cu catalyzes electrochemical CO₂ methanation. *Nano Lett* 21(17):7325–7331. <https://doi.org/10.1021/acs.nanolett.1c02502>
19. Ma W, He X, Wang W, Xie S, Zhang Q, Wang Y (2021) Electrocatalytic reduction of CO₂ and CO to multi-carbon compounds over Cu-based catalysts. *Chem Soc Rev* 50:12897–12914. <https://doi.org/10.1021/acs.materlett.1c00543>
20. Chen J, Feng J, Yang F, Aleisa R, Zhang Q, Yin Y (2019) Space-confined seeded growth of Cu nanorods with strong surface plasmon resonance for photothermal actuation. *Angew Chem Int Ed* 58(27):9275–9281. <https://doi.org/10.1002/anie.201904828>
21. Wei D, Tan Y, Wang Y, Kong T, Shen S, Mao S (2020) Function-switchable metal/semiconductor junction enables efficient photocatalytic overall water splitting with selective water oxidation products. *Sci Bull* 65(16):1389–1395. <https://doi.org/10.1016/j.scib.2020.04.042>
22. Duan S, Zhang I, Xie Z, Xu X (2020) Identification of water hexamer on Cu (111) surfaces. *J Am Chem Soc* 142(15):6902–6906. <https://doi.org/10.1021/jacs.0c01549>
23. Wu Y, Ma L, Song Z, Dong S, Guo Z, Wang J, Zhou Y (2023) Ordered mesoporous carbon encapsulated linear poly (ionic liquid) s enabling synergy effect of surface groups and ionic moieties for CO₂ fixation under mild conditions. *Carbon Neutrality* 2:1. <https://doi.org/10.1007/s43979-022-00041-5>
24. Ye Z, Lou M, Jia X, Shen J, Lu P, Huang H, Ye G, Yan B (2022) Reduction of the water wettability of Cu films deposited on liquid surfaces by thermal evaporation. *Colloid Surface A* 650:129569. <https://doi.org/10.1016/j.colsurfa.2022.129569>
25. Lee S, Kim D, Kim Y, Park S, Choi G, Kim Y, Kim H (2017) Droplet evaporation characteristics on transparent heaters with different wettabilities. *RSC Adv* 7:45274–45279. <https://doi.org/10.1039/C7RA08888D>
26. Pang Y, Zhang J, Ma R, Qu Z, Lee E, Luo T (2020) Solar-thermal water evaporation: a review. *ACS Energy Lett* 5(2):437–456. <https://doi.org/10.1021/acsenergylett.9b02611>
27. Alvarez-Paneque A, Rodríguez-González B, Pastoriza-Santos I, Liz-Marzán L (2013) Shape-templated growth of Au@Cu nanoparticles. *J Phys Chem C* 117(6):2474–2479. <https://doi.org/10.1021/jp3062724>
28. Zhu L, Gao M, Peh C, Ho G (2018) Solar-driven photothermal nanostructured materials designs and prerequisites for evaporation and catalysis applications. *Mater Horiz* 5(3):323–343. <https://doi.org/10.1039/C7MH01064H>
29. Halas N, Lal S, Chang W, Link S, Nordlander P (2011) Plasmons in strongly coupled metallic nanostructures. *Chem Rev* 111(6):3913–3961. <https://doi.org/10.1021/cr200061k>
30. Ghasemi H, Ni G, Marconnet A, Loomis J, Yerci S, Miljkovic N, Chen G (2014) Solar steam generation by heat localization. *Nat Commun* 5:4449. <https://doi.org/10.1039/C6TA01205A>
31. Manikandan S, Selvam C, Pavan Sai Praful P, Lamba R, Kaushik S, Zhao D, Yang R (2019) A novel technique to enhance thermal performance of a thermoelectric cooler using phase-change materials. *J Therm Anal Calorim* 140(3):1003–1014. <https://doi.org/10.1007/s10973-019-08353-y>
32. Pialago E, Yoo J, Zheng X, Kim B, Hong S, Kwon O, Park C (2020) Experimental investigation of the heat transfer performance of capillary-assisted horizontal evaporator tubes with sintered porous hydrophilic copper-carbon nanotube-titanium dioxide (Cu-CNT-TiO₂) composite coatings for adsorption chiller. *Int J Heat Mass Tran* 147:118958. <https://doi.org/10.1016/j.jheatmasstransfer.2020.119935>

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.